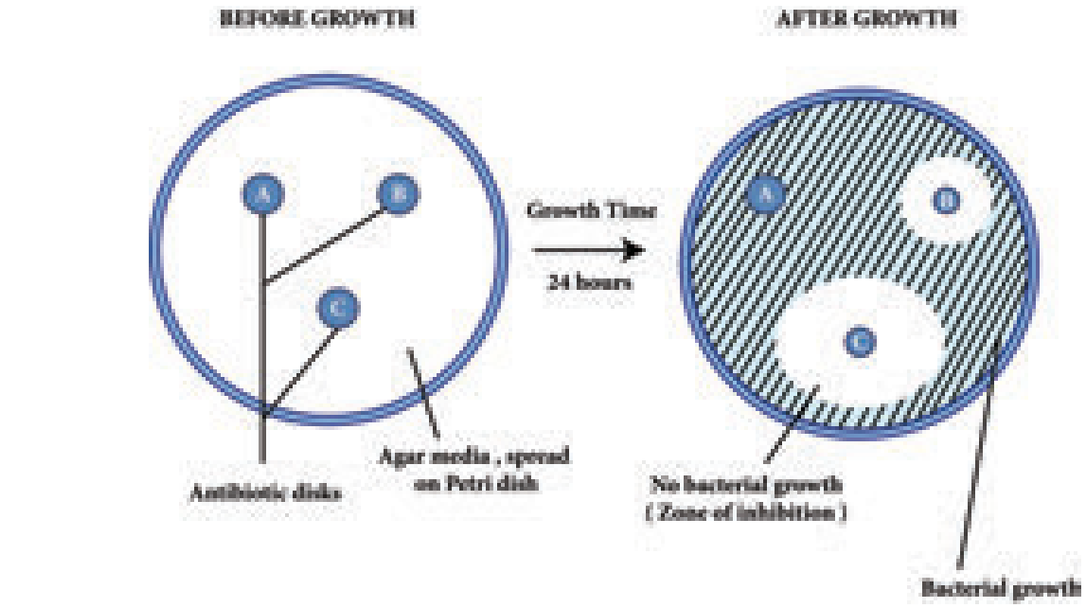


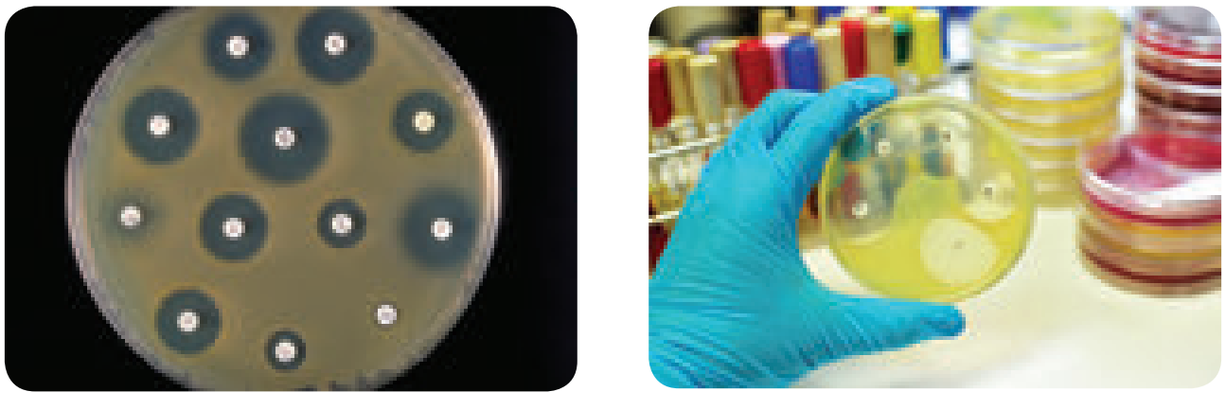


Figure 3.3: Microorganisms in Petri plate showing sensitivity and resistance towards antibiotics

and laboratory standards institute (CLSI) and must be accompanied by a rigorous quality assurance program including performance by certified and/or licensed personnel when the results are to be reported in clinical settings.

Tube Dilution Method

In this method, the test bacterium is inoculated into broth tubes containing serial dilution of antimicrobial agent. The inoculated cultures are incubated for a suiTable period of time (usually 24 hours), and the presence or absence of growth is determined by the turbidity in each tube. Tubes containing moderate to high concentrations of the antimicrobial agent would normally expected to have no growth.

Minimal Inhibitory Concentration (Mic) Test

The potency of an effective antimicrobial agent is expressed in terms of **minimal inhibitory concentration (MIC)**. It is the smallest amount of drug that will **inhibit** the growth of pathogen. The MIC is determined by serial dilutions of

TN_GOVT_XII_Micro_Biology_CH03.indd 40 15-04-2019 16:35:06